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2

CHE F311 Kinetics & Reactor Design

Objectives

- In this chapter we focus on
- ways of **obtaining and analyzing** reaction rate data to **obtain the rate law** for a specific reaction

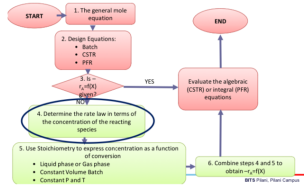
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Algorithm for Isothermal Reactor



4

DATA ACQUISITION

- Concentration-time measurements in a batch reactor and {transient operation}
- Concentration measurements in a differential reactor {steady-state operation}



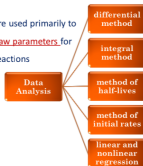
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BATCH REACTOR DATA- ANALYSIS

- Batch reactors are used primarily to determine **rate law parameters** for homogeneous reactions



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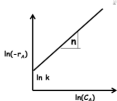
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Differential Method of analysis

$$-r_A = \frac{-dC_A}{dt} = k C_A^n$$

$$\ln(-r_A) = \ln(k) + n \ln(C_A)$$

If $-r_A = k C_A^a C_B^b$, how to use DM?



$$\ln(-r_A) = \ln(k) + a \ln(C_A) + b \ln(C_B)$$

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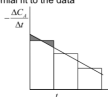
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Determine $(-dC_A/dt)$

Three ways

- Graphical differentiation
- Numerical differentiation formulas
- Differentiation of a polynomial fit to the data

1. Graphical



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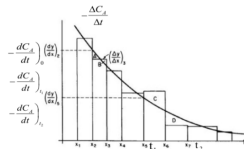
x_1	y_1	Δx	Δy	$\frac{\Delta x}{\Delta t}$	$\frac{dy}{dx}$
x_1	y_1				$\left(\frac{dy}{dx}\right)_1$
	$x_2 - x_1$	$y_2 - y_1$		$\left(\frac{\Delta x}{\Delta t}\right)_2$	
x_2	y_2				$\left(\frac{dy}{dx}\right)_2$
	$x_3 - x_2$	$y_3 - y_2$		$\left(\frac{\Delta x}{\Delta t}\right)_3$	
x_3	y_3				$\left(\frac{dy}{dx}\right)_3$
x_4	y_4			etc.	

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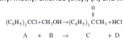
11

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Example

The reaction of triphenyl methyl chloride (trityl) (A) and methanol (B)



Time (min)	0	50	100	150	200	250	300
Concentration of A (mol/dm ³) $\times 10^3$	50	38	30.6	25.6	22.2	19.5	17.4

(At $t = 0$, $C_A = 0.05 \text{ M}$)

The initial concentration of methanol was 0.5 mol/dm^3 .

Part (i) Determine the reaction order with respect to triphenyl methyl chloride.

Part (ii) In a separate set of experiments, the reaction order wrt methanol was found to be first order. Determine the specific reaction rate constant

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t (min)	$C_A \times 10^3 \text{ (mol/dm}^3\text{)}$	$-\frac{dC_A}{dt} \times 10^3 \text{ (mol/dm}^3 \cdot \text{min)}$
0	50	
50	38	2.40
100	30.6	1.48
150	25.6	1.00
200	22.2	0.68
250	19.5	0.54
300	17.4	0.42

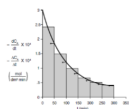
$$\frac{dC_A}{dt} = -\frac{C_A - C_{A,0}}{t - t_0} = -\left(\frac{38 - 50}{50 - 0}\right) \times 10^{-3} = 0.24 \times 10^{-3} = 2.4 \times 10^{-4} \text{ (mol/dm}^3 \cdot \text{min)}$$

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Graphical

$$-\frac{dC_A}{dt} \times 10^3 = 10,000$$

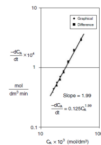
t (min)	(mol/dm ³ · min)
0	3.0
50	1.86
100	1.20
150	0.80
200	0.64
250	0.54
300	0.42

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Numerical method

Time (min)	t_0	t_1	t_2	t_3	t_4	t_5
Concentration (mol/dm ³)	C_{A0}	C_{A1}	C_{A2}	C_{A3}	C_{A4}	C_{A5}

$$\left(\frac{dC_A}{dt}\right)_{t_0} = \frac{-3C_{A0} + 4C_{A1} - C_{A2}}{2\Delta t}$$

$$\left(\frac{dC_A}{dt}\right)_{t_1} = \frac{1}{2\Delta t} [(C_{A1} + 1) - C_{A0}]$$

$$\left(\frac{dC_A}{dt}\right)_{t_5} = \frac{1}{2\Delta t} [C_{A5} - 4C_{A4} + 3C_{A3}]$$

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Finite difference method

Time (min)	0	50	100	150	200	250	300
Concentration of A (mol/dm ³) × 10 ³	50	38	30.6	25.6	22.2	19.5	17.4
(At t = 0, C _A = 0.05 M)							
$t \pm \Delta t$	$\left(\frac{dC_A}{dt}\right)_{t=0}$	$\frac{-3C_{A0} + 4C_{A1} - C_{A2}}{2\Delta t}$					
		$\frac{1}{2} \left[\frac{3(50) + 4(38) - 30.6}{100} \right]$					
		$= 2.38 \times 10^{-4} \text{ mol/dm}^3 \cdot \text{min}$					
		$\frac{dC_A}{dt} \times 10^3 = 2.38 \text{ mol/dm}^3 \cdot \text{min}$					
$t \pm 50$	$\left(\frac{dC_A}{dt}\right)_{t=50}$	$\frac{C_{A1} - C_{A0}}{\Delta t} = \frac{(38 - 50) \times 10^{-3}}{100}$					
		$= -1.24 \times 10^{-4} \text{ mol/dm}^3 \cdot \text{min}$					
$t \pm 300$	$\left(\frac{dC_A}{dt}\right)_{t=300}$	$\frac{C_{A5} - 4C_{A4} + 3C_{A3}}{2\Delta t} = \frac{17.2 - 4(19.5) + 3(17.4)}{100} \times 10^{-3}$					
		$= -8.36 \times 10^{-5} \text{ mol/dm}^3 \cdot \text{min}$					
		$\frac{dC_A}{dt} \times 10^3 \approx 10,000$					
		t (min)					
		(mol/dm ³ · min)					
		0	2.86				
		50	1.94				
		100	1.24				
		150	0.84				
		200	0.61				
		250	0.48				
		300	0.36				

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POLYNOMIAL FIT

- First fit the concentration-time data to an n^{th} -order polynomial

$$C_A = a_0 + a_1 t + a_2 t^2 + \dots + a_n t^n$$

$$\frac{dC_A}{dt} = a_1 + 2a_2 t + 3a_3 t^2 + \dots + n a_n t^{n-1}$$

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The difference between curve fitting and data interpolation

- In curve fitting the line need not pass through the data points but it has to in data interpolation

Curve fitting with polynomials is done in MATLAB with the `polyfit` function, which uses the least square method. Its syntax is:

$$p = \text{polyfit}(x, y, n)$$

p is the vector of the coefficients of the polynomial that fits the data

x is a vector of values of the independent variable

y is a vector of values of the dependent variable

n is the **degree** of the polynomial

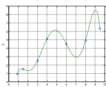
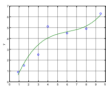
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```
>> x = [0.9 1.5 3 4 6 8 9.5]; y = [0.9 1.5 2.5 5.1 4.5 4.9 6.3];
>> p = polyfit(x,y,3);
>> xp = 0.9:0.1:9.5; yp = polyval(p,xp);
>> plot(x,y,'o',xp,yp); grid on; xlabel('x'); ylabel('y');
```



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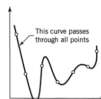
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Caution

if you use a computer to fit a polynomial to the data it could lead to disaster.



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Integral Method of analysis

- Guess the reaction order
- Integrate and Derive the equation
- Check whether the assumed order is correct or not by plotting the necessary graph

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Integral analysis

Mole Balances: $\frac{dN_A}{dt} = r_A V$ Rate Laws: $-r_A = k C_A^\alpha$

Combine:

$$-\frac{dC_A}{dt} = k C_A^\alpha$$

Zeroth order	First Order	Second Order
$\frac{dC_A}{dt} = r_A = -k$	$\frac{dC_A}{dt} = r_A = -k C_A$	$\frac{dC_A}{dt} = r_A = -k C_A^2$
at $t = 0$, $C_A = C_{A0}$	at $t = 0$, $C_A = C_{A0}$	at $t = 0$, $C_A = C_{A0}$
$\Rightarrow C_A = C_{A0} - kt$	$\Rightarrow \ln\left(\frac{C_{A0}}{C_A}\right) = kt$	$\Rightarrow \frac{1}{C_A} - \frac{1}{C_{A0}} = kt$

Eg: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

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Eg: H_2O_2 decomposition
Inversion of cane sugar

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Guess and check for $\alpha = 0, 1, 2$ and check against experimental plot.

$$\alpha = 0$$

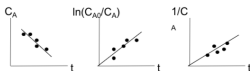
$$r_A = C_{A0} - kt$$

$$\alpha = 1$$

$$\ln\left(\frac{C_{A0}}{C_A}\right) = kt$$

$$\alpha = 2$$

$$\frac{1}{C_A} - \frac{1}{C_{A0}} = kt$$



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24

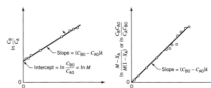
Irreversible Second-Order Reactions

Eg: $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$ • $A + B \rightarrow \text{Products}$

Caution

$$\ln \frac{1-X_A}{1-X_B} = \ln \frac{M-X_A}{M(1-X_A)} = \ln \frac{C_{B0}-C_A}{C_{B0}C_A} = \ln \frac{C_B}{MC_A} = \ln \frac{C_B}{C_A} - \ln M$$

$$= C_{A0}(M-1)kt = (C_{B0}-C_{A0})kt, \quad M \neq 1$$



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Caution

- If the order of the reaction is **first order with respect to both A and B**, hence second order overall

$$-r_A = -\frac{dC_A}{dt} = kC_A C_B = kC_{A0}^2(1 - X_A)(M - 2X_A)$$

$$\ln \frac{C_B C_{A0}}{C_{B0} C_A} = \ln \frac{M - 2X_A}{M(1 - X_A)} = kC_{A0}(M - 2)kt, \quad M \neq 2$$

When a stoichiometric reactant ratio is used

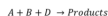
$$\frac{1}{C_A} - \frac{1}{C_{A0}} = \frac{1}{C_{A0}} \frac{X_A}{1 - X_A} = 2kt, \quad M = 2$$

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3rd Order reaction

$$\text{Rate Law: } -r_A = -\frac{dC_A}{dt} = kC_A C_B C_D$$

$$\text{Rate Law in terms of } X: C_{A0} \frac{dX_A}{dt} = kC_{A0}^3(1 - X_A) \left(\frac{C_{B0}}{C_{A0}} - X_A \right) \left(\frac{C_{D0}}{C_{A0}} - X_A \right)$$

$$\frac{1}{(C_{A0} - C_{B0})(C_{A0} - C_{D0})} \ln \frac{C_{D0}}{C_A} + \frac{1}{(C_{B0} - C_{D0})(C_{B0} - C_{A0})} \ln \frac{C_{B0}}{C_B} + \frac{1}{(C_{D0} - C_{A0})(C_{D0} - C_{B0})} \ln \frac{C_{D0}}{C_D} = kt$$

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3rd Order reaction

$$\text{Rate Law: } -r_A = -\frac{dC_A}{dt} = kC_A C_B^2$$

$$\text{In terms of conversion } \frac{dX_A}{dt} = kC_{A0}^3(1 - X_A)(M - 2X_A)^2$$

$$\frac{(2C_{A0} - C_{B0})(C_{B0} - C_B)}{C_{B0} C_B} + \ln \frac{C_{A0} C_B}{C_A C_{B0}} = (2C_{A0} - C_{B0})^2 kt, \quad M \neq 2$$

$$\frac{1}{C_A^2} - \frac{1}{C_{A0}^2} = 8kt, \quad M = 2$$

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nth order reaction

$$\text{For a } n^{\text{th}} \text{ order reaction: } -r_A = -\frac{dC_A}{dt} = kC_A^n$$

$$\text{On integration: } C_A^{1-n} - C_{A0}^{1-n} = (n-1)kt, \quad n \neq 1$$

The order "**n**" cannot be found explicitly from Eq., so a trial-and-error solution must be made.

For $n > 1$, never go to completion in finite time

$$\text{For } n < 1, C_A = 0 \text{ when } t = \frac{C_{A0}^{1-n}}{(1-n)k}$$

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Autocatalytic Reactions

One of the product from the reaction acts as a catalyst.



$$-r_A = -\frac{dC_A}{dt} = kC_A C_R$$

At any time t : $C_0 = C_A + C_R = C_{A0} + C_{R0} = \text{constant}$

$$-r_A = -\frac{dC_A}{dt} = kC_A(C_0 - C_A)$$

$$-\frac{dC_A}{C_A(C_0 - C_A)} = -\frac{1}{C_0} \left(\frac{dC_A}{C_A} + \frac{dC_A}{C_0 - C_A} \right) = k dt$$

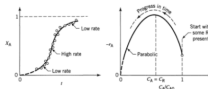
$$\ln \frac{C_{A0}(C_0 - C_A)}{C_A(C_0 - C_{A0})} = \ln \frac{C_{A0}C_{R0}}{C_A C_{A0}} = C_0 k t = (C_{A0} + C_{R0}) k t$$

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Autocatalytic Reactions



$$M = C_{A0}C_{A0}$$

$$\ln \frac{M + X_A}{M(1 - X_A)} = C_{A0}(M + 1)kt = (C_{A0} + C_{R0})kt$$



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Series Reactions



$$r_A = \frac{dC_A}{dt} = -k_1 C_A$$

$$r_R = \frac{dC_R}{dt} = k_1 C_A - k_2 C_R$$

$$r_S = \frac{dC_S}{dt} = k_2 C_R$$

$$\frac{dC_R}{dt} + k_2 C_R = k_1 C_{A0} e^{-k_1 t}$$

$$C_R = C_{A0} k_1 \left(\frac{e^{-k_1 t}}{k_2 - k_1} + \frac{e^{-k_2 t}}{k_1 - k_2} \right)$$

$$C_{A0} = C_A + C_R + C_S \quad C_S = C_{A0} \left(1 + \frac{k_2}{k_1 - k_2} e^{-k_1 t} + \frac{k_1}{k_2 - k_1} e^{-k_2 t} \right)$$

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Reactions of Shifting order

➤ In searching for a kinetic equation it may be found that the data are well fitted by one reaction order at high concentrations but by another order at low concentrations



$$-r_A = -\frac{dC_A}{dt} = \frac{k_1 C_A}{1 + k_2 C_A}$$

Case 1: At high $C_A \rightarrow$ reaction is of zero order with rate constant k_1/k_2

Case 2: At low $C_A \rightarrow$ reaction is of first order with rate constant k_1

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$$-r_A = -\frac{dC_A}{dt} = \frac{k_1 C_A}{1 + k_2 C_A}$$

By applying the integral method

$$\ln \frac{C_{A0}}{C_A} + k_2 (C_{A0} - C_A) = k_1 t$$

Rearrange:

$$\frac{C_{A0} - C_A}{\ln(C_{A0}/C_A)} = -\frac{1}{k_2} + \frac{k_1}{k_2} \left(\frac{t}{\ln(C_{A0}/C_A)} \right)$$

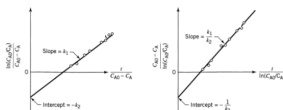
$$\frac{\ln(C_{A0}/C_A)}{C_{A0} - C_A} = -k_2 + \frac{k_1}{C_{A0} - C_A}$$

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Eg: Diethylsilane pyrolysis to silicon carbide deposition

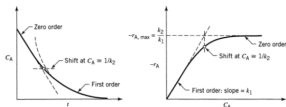
Ref: [Journal of Materials Science Letters 13\(17\):1237-1239](#)

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Eg: Diethylsilane pyrolysis to silicon carbide deposition

Ref: [Journal of Materials Science Letters 13\(17\):1237-1239](#)

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First order reversible

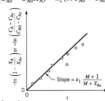


$$\frac{dC_A}{dt} = -\frac{dC_A}{dt} = C_{A0} \frac{dX_A}{dt} = k_1 C_A - k_2 C_R = k_1 (C_{A0} - C_{A0} X_A) - k_2 (MC_{A0} + C_{A0} X_A)$$

$$K_C = \frac{C_R}{C_A} = \frac{M + X_{Ae}}{1 - X_{Ae}}$$

$$\frac{dX_A}{dt} = \frac{k_1 (M + 1)}{M + X_{Ae}} (X_{Ae} - X_A)$$

$$-\ln \left(1 - \frac{X_A}{X_{Ae}} \right) = -\ln \frac{C_A - C_{Ae}}{C_{A0} - C_{Ae}} = \frac{M + 1}{M + X_{Ae}} k_1 t$$



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2nd order reversible



With $C_{A0} = C_{B0}$ and $C_{R0} = C_{S0} = 0$



$$\ln \frac{X_A - (2X_{Ae} - 1)X_A}{X_{Ae} - X_A} = 2k_1 \left(\frac{1}{X_{Ae}} - 1 \right) C_{A0} t$$

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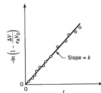
Variable volume

For first order:



$$-r_A = kC_A = kC_{A0} \left(\frac{1 - X_A}{1 + \epsilon_A X_A} \right)$$

$$-\ln \left(1 - \frac{\Delta V}{\epsilon_A V_0} \right) = k t, \quad \Delta V = V - V_0$$



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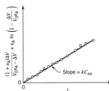
Variable volume

For second order:

- For a bimolecular-type second-order reaction, with $C_{A0} = C_{B0}$ and the rate is given by

$$-r_A = kC_A^2 = kC_{A0}^2 \left(\frac{1 - X_A}{1 + \epsilon_A X_A} \right)^2$$

$$\frac{(1 + \epsilon_A) \Delta V}{V_0 \epsilon_A - \Delta V} + \epsilon_A \ln \left(1 - \frac{\Delta V}{\epsilon_A V_0} \right) = k C_{A0} t$$



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Integral vs Differential

Integral Method	Differential Method
- Easy to use and is recommended for testing specific mechanism - Require small amount of data - Involves trial and error - Cannot be used for fractional orders - Very accurate	- Useful in complicated cases - Require large and more accurate data - No trial and error - Can be used for fractional orders - Less accurate

Generally Integral Method is attempted first and if not successful, the differential method is used.

41

Non linear regression

$$\frac{dC_A}{dt} = -kC_A^\alpha$$

$$t = 0 \quad C_A = C_{A0}$$

$$C_{A0}^{1-\alpha} - C_A^{1-\alpha} = (1-\alpha)kt$$

$$C_{Ac} = C_A = [C_{A0}^{1-\alpha} - (1-\alpha)kt]^{1/(1-\alpha)}$$

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42

We want to find the parameter values (α, k) for which the sum of the squares of the differences, the measured concentration (C_m), and the calculated concentration (C_c) is a minimum.

$$\sigma^2 = \sum_{j=1}^N \frac{(C_m - C_c)^2}{N-K} = \frac{S^2}{N-K}$$

K is number of parameters to be determined

That is, we want σ^2 to be a minimum.

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43

Minimize the sum of squares of differences between the measured (C_m) and calculated (C_c) concentrations.

That is, for N data points,

$$S^2 = \sum_{j=1}^N (C_{Am} - C_{Ac})^2 = \sum_{j=1}^N [C_{Am} - [C_{A0}^{1-\alpha} - (1-\alpha)kt_j]^{1/(1-\alpha)}]^2$$

Similarly one can calculate the time at a specified concentration, t_c

$$t_c = \frac{C_{A0}^{1-\alpha} - C_A^{1-\alpha}}{k(1-\alpha)}$$

$$S^2 = \sum_{j=1}^N (t_{Am} - t_{cj})^2 = \sum_{j=1}^N \left[t_{Am} - \frac{C_{A0}^{1-\alpha} - C_{Aj}^{1-\alpha}}{k(1-\alpha)} \right]^2$$

and compare it with the measured time, t_{Am} at that same concentration.

That is, we find the values of k and α that minimize:

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44

Original Data	Guess 1		Guess 2		Guess 3		Guess 4	
	$\alpha = 3$ $k' = 5$		$\alpha = 2$ $k' = 5$		$\alpha = 2$ $k' = 0.2$		$\alpha = 2$ $k' = 0.1$	
t (min)	$C_A \times 10^3$ (mol/dm ³)	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²	t_c ($t_{Am} - t_c$) ²
1	0	50	0	0	0	0	0	0
2	50	38	29.2	433	1.26	2,375	31.6	339
3	100	30.6	66.7	1,109	2.5	9,499	63.4	1,340
4	200	22.2	163	1,375	5.0	34,622	125.2	5,591
		$S^2 = 2916$		$S^2 = 49,895$		$S^2 = 7270$		$S^2 = 3433$

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45

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$$s^2 = \sum_{i=1}^N (C_{Aei} - C_{Aci})^2 \approx \sum_{i=1}^N \left(C_{Aei} - [C_{A0}^{1-\alpha} - (1-\alpha)k t_i]^{1/(1-\alpha)} \right)^2$$

We find the values of alpha and k which minimize s^2

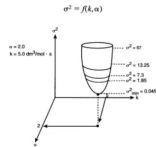


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46

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47

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```
function t = ODEfun(yo, Ca)
a=yo(1);
k=yo(2);
t=(.05*(1-a)*Ca^(1-a))/(k*(1-a)); % Model
```

```
clear
clc
close all
t=[0:50;100;150;200;250;300]; % Experimental time(min) value
Ca=[0.05;0.038;0.0306;0.0256;0.0222;0.0195;0.0174]; % Experimental Ca(mol/dm^3) value
yo=[3;0.1]; % initial guess vector for a and k
yfinal=nlmfit(Ca,t,@ODEfun,yo);
a=yfinal(1);
k=yfinal(2);
disp('The final value of alpha and k(dm^3/mol.min) are')
a
k
a= 2.04
k= 0.147
```

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CHE 7511 Kinetics & Reactor Design



2nd Order RXN

```
function t = ODEfun1(yo, Ca)
k=yo(1);
a=2; % 2nd order reaction
t=(.05*(1-a)*Ca^(1-a))/(k*(1-a)); % Model
```

```
clear
clc
close all
t=[0:50;100;150;200;250;300]; % Experimental time(min) value
Ca=[0.05;0.038;0.0306;0.0256;0.0222;0.0195;0.0174]; % Experimental Ca(mol/dm^3) value
yo=[0.1]; % initial guess vector for k
yfinal=nlmfit(Ca,t,@ODEfun1,yo);
k=yfinal(1);
disp('The final value of k(dm^3/mol.min) is')
k
k= 0.125
```

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49

CHE 3202 Kinetics & Reactor Design

Method of Half lives

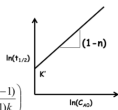
$$\frac{1}{C_A^{n-1}} - \frac{1}{C_{A0}^{n-1}} = (n-1)kt \quad (n \neq 1)$$

$$At \ t = t_{1/2}, \ C_A = C_{A0}/2$$

$$t_{1/2} = \frac{C_{A0}^{1-n} (2^{n-1} - 1)}{(n-1)k}$$

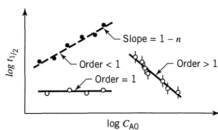
$$\ln(t_{1/2}) = (1-n) \ln(C_{A0}) + \ln\left(\frac{2^{n-1} - 1}{(n-1)k}\right)$$

Require multiple experiments

For 1st Order reactions: $t_{1/2} = \ln(2)/k$ does not depend on C_{A0} 

CHE 3202 Kinetics & Reactor Design

Method of Half lives



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56

CHE 3202 Kinetics & Reactor Design

- A plug flow reactor (2 m^3) processes an aqueous feed (100 liter/min) containing reactant A ($C_{A0} = 100 \text{ mmol/liter}$). This reaction is reversible and represented by $A \rightleftharpoons B$

First find the equilibrium conversion and then find the actual conversion of A in the reactor.

b. CSTR

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CHE 3202 Kinetics & Reactor Design

Batch Reactor- Constant Volume- Total pressure

- Find the rate that represents the decomposition of A, $A \rightarrow R + 2S$, from the following pressure data. Assume pure A present initially.

Time (min)	0	2.5	5	10	15	20
Tot. pressure (mm Hg)	7.5	10.5	12.5	15.8	17.9	19.4

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BATCH REACTOR DATA- ANALYSIS



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64

DIFFERENTIAL REACTORS

- Data acquisition using the method of initial rates and a differential reactor are similar.
- The rate of reaction is determined for a specified number of predetermined initial or entering reactant concentrations.
- A differential reactor is normally used to determine the rate of reaction as a function of either concentration or partial pressure.

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- Differential reactor is assumed to be *gradient less*, the design equation will be similar to the CSTR design equation.
- If the weight of catalyst, ΔW , is known, the rate of reaction per unit mass of catalyst, $-r_A'$ can be calculated.



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66

$$\left[\begin{array}{c} \text{Flow} \\ \text{rate} \\ \text{in} \end{array} \right] - \left[\begin{array}{c} \text{Flow} \\ \text{rate} \\ \text{out} \end{array} \right] + \left[\begin{array}{c} \text{Rate of} \\ \text{generation} \end{array} \right] = \left[\begin{array}{c} \text{Rate of} \\ \text{accumulation} \end{array} \right]$$

$$-r_A' = \frac{F_{A0} - F_{A2}}{\Delta W} \quad -r_A' = \frac{v_0 C_{A0} - v C_{A2}}{\Delta W}$$

$$-r_A' = \frac{F_{A0} V - F_P}{\Delta W} \quad -r_A' = \frac{v_0 (C_{A0} - C_{A2})}{\Delta W} = \frac{v_0 C_P}{\Delta W}$$

$-r_A'$ can be determined by measuring the product concentration, C_P

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CHE 3321 Kinetics & Reactor Design



- By using very little catalyst and large volumetric flow rates, the concentration difference, ($C_{Ab} - C_{Ar}$) can be made quite small.

- The rate of reaction determined from previous equation, can be obtained as a function of the reactant concentration in the catalyst bed, C_{Ab}

$$-r'_A = -r'_A(C_{Ab})$$

$$C_{Ab} = \frac{C_{Ar} + C_{Ar'}}{2}$$

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68

CHE 3321 Kinetics & Reactor Design



- However, since very little reaction takes place within the bed

$$C_{Ab} \approx C_{Ar}$$

$$-r'_A = -r'_A(C_{Ar})$$

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69

CHE 3321 Kinetics & Reactor Design



- Find the overall order of the irreversible reaction

$2H_2 + 2NO \rightarrow N_2 + 2H_2O$. From the following constant-volume data using equimolar amounts of hydrogen and nitric oxide:

Total pressure, mm Hg 200 240 280 320 360

Half-life, sec 265 186 115 104 67

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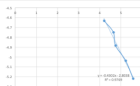
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70

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P (mm Hg)	T _{1/2}	P (atm)	CAO	lnCAO	lnT _{1/2}
200	265	0.263158	0.005417	-5.2182	5.57973
240	186	0.315789	0.0065	-5.03588	5.225747
280	115	0.368421	0.007584	-4.88173	4.746932
320	104	0.421053	0.008687	-4.7482	4.643791
360	67	0.473684	0.009751	-4.63042	4.204659



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71



The first-order reversible liquid reaction



takes place in a batch reactor. After 8 minutes, conversion of A is 33.3% while equilibrium conversion is 66.7%. Find the rate equation for this reaction.

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72



Find the conversion after 1 hour in a batch reactor for



Find the first-order rate constant for the disappearance of A in the gas reaction $2A \rightarrow R$ if, on holding the pressure constant, the volume of the reaction mixture, starting with 80% A, decreases by 20% in 3 min.

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73



L 5.19

Pure gaseous A at about 3 atm and 30°C (120 mmol/liter) is fed into a 1-liter mixed flow reactor at various flow rates. There it decomposes, and the exit concentration of A is measured for each flow rate. From the following data find a rate equation to represent the kinetics of the decomposition of A. Assume that reactant A alone affects the rate.

v_D , liter/min	0.06	0.48	1.5	8.1
C_A , mmol/liter	30	60	80	105



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74



F 5.8

The following data were reported [C. N. Hinshelwood and P. J. Ackley, *Proc. R. Soc. (Lond.)*, A143, 215] for a gas-phase constant-volume decomposition of dimethyl ether at 504°C in a batch reactor. Initially, only $(CH_3)_2O$ was present.

Time (s)	390	777	1195	3155	∞
Total Pressure (mmHg)	408	488	562	799	931

- (a) Why do you think the total pressure measurement at $t = 0$ is missing? Can you estimate it?
- (b) Assuming that the reaction



- is irreversible and goes to completion, determine the reaction order and specific reaction rate k .
- (c) What experimental conditions would you suggest if you were to obtain more data?
- (d) How would the data and your answers change if the reaction were run at a higher or lower temperature?

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75



L 3.25

The following data are obtained at 0°C in a constant-volume batch reactor using pure gaseous A:

Time, min	0	2	4	6	8	10	12	14	∞
Partial pressure of A, mm	760	600	475	390	320	275	240	215	150

The stoichiometry of the decomposition is $A \rightarrow 2.5R$. Find a rate equation which satisfactorily represents this decomposition.



L 5.2

Pure gaseous reactant A ($C_{A0} = 100$ millimol/liter) is fed at a steady rate into a mixed flow reactor ($V = 0.1$ liter) where it dimerizes ($2A \rightarrow R$). For different gas feed rates the following data are obtained:

Run number	1	2	3	4
v_0 , liter/hr	10.0	3.0	1.2	0.5
C_{AR} , millimol/liter	85.7	66.7	50	33.4

Find a rate equation for this reaction.



A small reaction bomb fitted with a sensitive pressure-measuring device is flushed out and then filled with pure reactant A at 1-atm pressure. The operation is carried out at 25°C, a temperature low enough that the reaction does not proceed to any appreciable extent. The temperature is then raised as rapidly as possible to 100°C by plunging the bomb into boiling water, and the readings in Table P3.21 are obtained. The stoichiometry of the reaction is $2A \rightarrow B$, and after leaving the bomb in the bath over the weekend the contents are analyzed for A; none can be found. Find a rate equation in units of moles, liters, and minutes which will satisfactorily fit the data.



T , min	π , atm	T , min	π , atm
1	1.14	7	0.850
2	1.04	8	0.832
3	0.982	9	0.815
4	0.940	10	0.800
5	0.905	15	0.754
6	0.870	20	0.728



- Raju likes to play the gambling tables just to get relaxed. He never expects to win and he doesn't, so he selects games in which losses are a given small fraction of the money bet. He plays steadily without a break, *and the sizes of his bets are proportional to the money he has*. If at gambling table-1 it takes him four hours to lose of half of his money and gambling table-2 it takes two hours to lose half of his money. How long can he play at both the tables simultaneously if he starts with Rs. 1000 and quits when he has Rs. 10 left?

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80
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- A homogeneous reaction $A \rightarrow S$, $-r_A = kC_A^2$ takes place in a mixed flow reactor and results in 50% conversion.
 - Find the conversion if this reactor is replaced by another MFR having a volume of 6 times that of the original reactor- all else remains unchanged.
 - Find the conversion X_A if the MFR is replaced by a PFR of the same size- all else remains unchanged.

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81
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Consider a feed $C_{A0} = 100$, $C_{B0} = 200$, $C_0 = 100$ to a steady-flow reactor. The isothermal gas-phase reaction is



If $C_A = 40$ at the reactor exit, what is C_B , X_A , and X_B there?

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82
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Given a gaseous feed, $T_0 = 1000$ K, $\pi_0 = 5$ atm, $C_{A0} = 100$, $C_{B0} = 200$, $A + B \rightarrow 5R$, $T = 400$ K, $\pi = 4$ atm, $C_A = 20$. Find X_A , X_B , C_B .

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83
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CHE 5201 Kinetics & Reactor Design



At room temperature sucrose is hydrolyzed by the catalytic action of the enzyme sucrase as follows:



Starting with a sucrose concentration $C_{A0} = 1.0$ millimol/liter and an enzyme concentration $C_{E0} = 0.01$ millimol/liter, the following kinetic data are obtained in a batch reactor (concentrations calculated from optical rotation measurements):

C_A , millimol/liter	0.84	0.68	0.53	0.38	0.27	0.18	0.09	0.04	0.018	0.006	0.0025
t , hr	1	2	3	4	5	6	7	8	9	10	11

Determine whether these data can be reasonably fitted by a kinetic equation of the Michaelis-Menten type, or

$$-r_A = \frac{k_2 C_A C_E}{C_M + C_A} \quad \text{where } C_M = \text{Michaelis constant}$$

1. If the fit is reasonable, evaluate the constants k_2 and C_M . Solve by the integral method.

Plot, Print, Compare